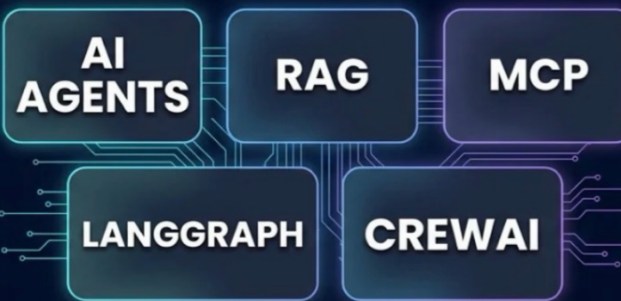


VIDEO 2



COMPLETE SET-UP ✓



codestorywithmiK



CSwithMIK



codestorywithMIK

Set-Up



VS Code

- You can choose your favorite IDE
- Install python extension in your IDE
- Why? Free, lightweight,
crazy extension ecosystem, terminal
etc.
- Alternatives - Pycharm, Cweser



Python3 latest stable version

- brew install python3 (Mac/Linux) ↓ ↓
- winget install Python.Python.3.12 (windows)
- why? Standard language for AI/ML.
OpenAI SDK, Ollama's Package,
Langchain, PyTorch, etc. are
all Python-first.
- Alternative - JS/Go etc.



(Deep Dive)

- Free, open source.
- Ollama.com download (Linux/MAC/windows)
- why? Easiest way to run LLM models locally
 - Free, No cloud API costs
 - No internet dependency once downloaded
 - Private to you.
 - Beginner Friendly.





- we will download an LLM model.
- Do some post-mortem.



- we will set up a simple project folder } *vscode*
(feel free to do it in your style)



- we will install relevant Python Packages
+ we know "WHY" behind everything }



- we will learn whatever is required in Python for this work. No need to do complete python before-hand as of now. }



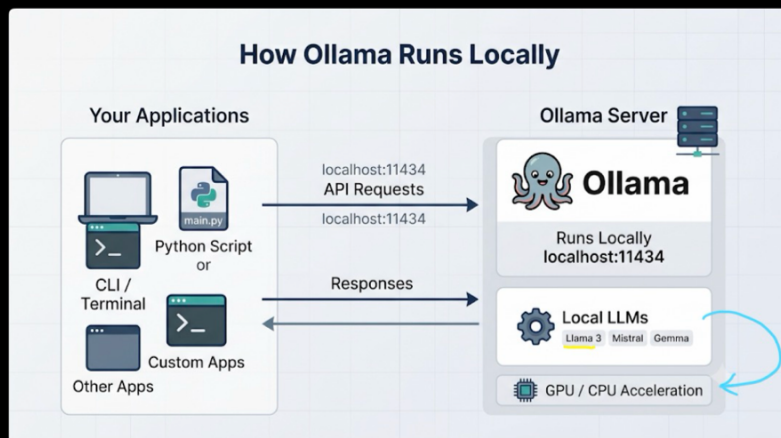
I'll ensure, I cover everything needed for even a beginner to understand it in simplest way possible.

Year.
2023



- Tool — Runs LLM in your laptop.
- Download model — Runs 100% offline — free
- Runs a local server on your machine

Ollama Serve



loading it
into Machine's
Memory/GPU

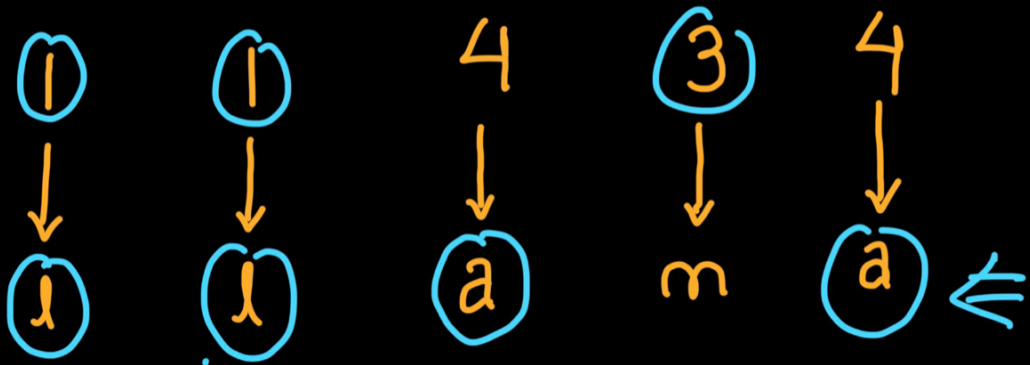
* Runs a background server locally

http://localhost 11434

* Exposes an openAI-compatible REST API

* You can talk to the LLM

* Port : 11434



Install

macOS

Linux

Windows

```
curl -fsSL https://ollama.com/install.sh | sh
```



Install

macOS

Linux

Windows

```
curl -fsSL https://ollama.com/install.sh | sh
```



Install

macOS

Linux

Windows

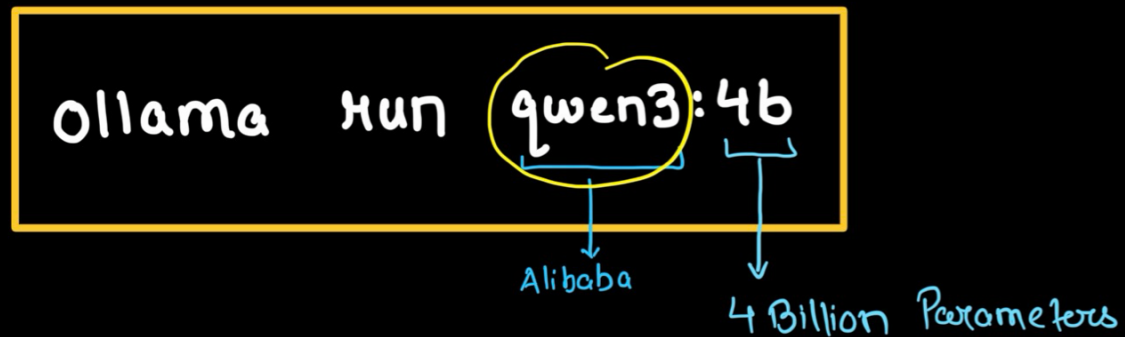
```
irm https://ollama.com/install.ps1 | iex
```



Run it:

ollama serve

Download Model: family:size



→ This model learned during its training.

"We will choose smaller sized for testing
& learning purposes"

Post-mortem...

- ① when you download a model, what do you actually get in your machine ??? → qwen3:4b

```
MIK ~  
ls ~/.ollama/ cache/ history id_ed25519 id_ed25519.pub models/
```

```
MIK ~  
ls ~/.ollama/models/ blobs/ manifests/ metadata/
```

Why files stored as Hash ??? (digest)

```
MIK ~/.ollama/models/blobs
ls
sha256-2d54db2b9bb29ce7db54fea63a891f5859603813c555b1f88b5e0994652897f9
sha256-3e4cb14174460404e7a233e531675303b2fbf7749c02f91864fe311ab6344e4f
sha256-cff3f395ef3756ab63e58b0ad1b32bb6f802905cae1472e6a12034e4246fbbdb
sha256-d18a5cc71b84bc4af394a31116bd3932b42241de70c77d2b76d69a314ec8aa12
sha256-e18a783aae5525fd2852fc94c985541a77e791e034abc2d3056474d59de336fc
MIK ~/.ollama/models/blobs
du -h sha256-2d54db2b9bb29ce7db54fea63a891f5859603813c555b1f88b5e0994652897f9
4.0K sha256-2d54db2b9bb29ce7db54fea63a891f5859603813c555b1f88b5e0994652897f9
MIK ~/.ollama/models/blobs
du -h sha256-3e4cb14174460404e7a233e531675303b2fbf7749c02f91864fe311ab6344e4f
2.3G sha256-3e4cb14174460404e7a233e531675303b2fbf7749c02f91864fe311ab6344e4f
MIK ~/.ollama/models/blobs
```

1st Reason

ollama $\xrightarrow{\text{SHA}}$ hash

textfile $\xrightarrow{\text{SHA}}$ xyz
textfile $\xrightarrow{\text{SHA}}$ xyz

} Hash value.

2nd Reason

qwen3:4b

models/sha-22abc-

qwen3:1.7b

models/sha-21abc

Docker / Git / Huggingface.

→ "CAS - Content - Addressable Storage"

file1.txt

file2.txt

SHA 256: --

models/

[SHA.....]